

MOHAMED MOUSTAFA DAWOUD

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RESEARCH INTERESTS

My research focuses on **security, privacy, and responsible AI**, with a **human-centered** emphasis on how people understand, experience, and govern intelligent systems.

Human-Centered Security & Privacy • AI Governance & Accountable Deployment • Human-AI Interaction, Trust & Misuse • Empirical & Data-Driven Auditing • Internet Measurements • Privacy Engineering

EDUCATION

University of California, Santa Cruz

2025 – present

Ph.D. in Computer Science & Engineering

Advisor: Prof. Ramakrishnan (Ram) Sundara Raman.

Focus: Human-Centered Security, Privacy, and Responsible AI.

German International University, Cairo, Egypt

2021 – 2024

B.Sc. in Informatics & Computer Science

Full merit scholarship (top-ranked national students); 18th nationwide in Thanawya Amma (High School) Mathematics (~600,000 students), 408/410 (99.51%).

PUBLICATIONS

From Underground to Mainstream Marketplaces: Measuring AI-Enabled NSFW Deepfakes on Fiverr, *Symposium on Usable Security and Privacy (USEC), co-located with NDSS* 2026
Mohamed Moustafa Dawoud, Alejandro Cuevas, and Ram Sundara Raman.

Vendor communication themes in darknet Ransomware-as-a-Service (RaaS) advertisements, Computers in Human Behavior 2025
Taylor Fisher, Zacharias Pieri, C. Jordan Howell, Roberta O'Malley, Lauren Tremblay, and Mohamed Dawoud.

DVa: Extracting Victims and Abuse Vectors from Android Accessibility Malware, *33rd USENIX Security Symposium (USENIX Security 24)* 2024
Haichuan Xu, Mingxuan Yao, Runze Zhang, Mohamed Moustafa Dawoud, Jeman Park, and Brendan Saltaformaggio.

Social Engineering and Technical Security Fusion, *International Telecommunications Conference (ITC-Egypt)* 2022
Wassim Alexan, Eyad Mamdouh, Mohamed ElBeltagy, Ahmed Ashraf, Mohamed Moustafa, and Hashem Al-Qurashi.

Image Encryption Through Rössler System, PRNG S-Box and Recamán's Sequence, IEEE 12th Annual Computing and Communication Workshop and Conference (CCWC) 2022
Mohamed ElBeltagy, Wassim Alexan, Abdelrahman Elkhamry, **Mohamed Moustafa**, and Hisham H. Hussein.

IoT Security: SHA3-512, AES-256, RSA and LSB Steganography, 8th NAFOSTED Conference on Information and Computer Science (NICS) 2021
Wassim Alexan, Ahmed Ashraf, Eyad Mamdouh, Sarah Mohamed, and **Mohamed Moustafa**.

ONGOING PROJECTS

AI Privacy & Regulation Study

Active

Qualitative study with UC Santa Cruz (RANDLab) and Stanford University examining how practitioners navigate privacy challenges and emerging regulations (EU AI Act, CCPA, GDPR) when building and deploying AI products. Conducting confidential interviews with professionals across privacy, compliance, engineering, and risk roles. IRB-approved (HS-FY2026-108).

RESEARCH & ACADEMIC EXPERIENCE

University of California, Santa Cruz

Jun 2025 – present

Graduate Student Researcher

- Conduct research on human-centered security, privacy, and AI governance under Prof. Ramakrishnan (Ram) Sundara Raman, focusing on how organizations deploy, regulate, and manage risks in emerging AI systems.
- Lead the **AI Privacy & Regulation Study** (with Stanford University): a qualitative study examining how practitioners navigate privacy challenges and emerging regulations (EU AI Act, CCPA, GDPR) when building and deploying AI products, through confidential interviews with industry professionals across privacy, compliance, engineering, and risk roles (IRB HS-FY2026-108).
- Lead mixed-methods research on AI-enabled abuse ecosystems, including large-scale measurements of AI abuse-as-a-service on gig marketplaces and expert interviews with industry practitioners.
- Develop structured LLM-based evaluation frameworks for analyzing organizational AI and privacy policies and identifying gaps between stated governance expectations and real-world practices.
- Designed a large-scale auditing pipeline integrating browser-based tracking measurements with rubric-guided LLM privacy-policy analysis to evaluate sector-wide compliance with modern privacy regulations (CCPA).

Dartmouth College

Jan – Jun 2025

Lead Graduate Teaching Assistant — COSC 55: Security & Privacy

- Served as Lead Graduate Teaching Assistant for COSC 55: Security & Privacy (Prof. Sami Saydjari), supporting a class of 28 students in an advanced undergraduate course on computer and network security.
- Designed and implemented a new AI-integrated teaching model that embraced tools such as ChatGPT and GitHub Copilot to foster critical thinking, explanation, and reflection rather than rote code production.
- Created a new series of labs emphasizing system design freedom, oral reasoning, and justification of design choices; guided weekly one-on-one and team check-ins to assess conceptual understanding.
- Conducted all grading, oral evaluations, and rubric design; emphasized rigorous yet supportive assessment through structured oral defenses.

- Mentored students on trade-offs in assurance, usability, and privacy engineering—culminating in innovative final projects (e.g., a privacy-preserving, role-based messaging system with blind signature tokens).
- Collaborated closely with the course instructor to co-develop teaching materials, improve pedagogical alignment, and evaluate how AI tools reshape student learning in computer security.

Mohamed bin Zayed University of AI (MBZUAI)

Sep – Dec 2024

Research Associate

- Conducted research on combating deepfakes and promoting the responsible use of AI through a human-centered lens.
- Investigated how users form mental models and trust judgments when interacting with AI-generated deepfakes across visual and textual media.
- Developed prototype systems and experimental frameworks for detecting AI-generated content and explaining authenticity decisions to end users.
- Explored strategies for improving human–AI collaboration in misinformation detection and enhancing transparency and interpretability in AI-driven systems.

Georgia Institute of Technology

Oct 2023 – Feb 2024

Research Intern

- Systems/network security research under Prof. Brendan D. Saltaformaggio; contributions to work that appeared later at USENIX Security 2024.
- Investigated the potential of large language models to assist static analysis of Android malware.
- Designed and executed a large-scale experiment using a manually curated dataset of benign and malicious Android malware static code collected from the wild.
- Demonstrated strong LLM capabilities for reasoning about malicious behaviors and summarizing complex code patterns; results submitted to the DIMVA conference.

University of South Florida

Jan – Aug 2023

Undergraduate Research Assistant (Remote)

- Worked with Prof. C. Jordan Howell on cybercrime and underground economies, with a focus on Ransomware-as-a-Service (RaaS) marketplaces.
- Led the threat-intelligence component of the project, including data collection and analysis of darknet RaaS vendor advertisements.
- Contributed to the analysis that uncovered core communication patterns (e.g., victimization, persuasion strategies) used to market pre-built ransomware offerings.
- This work appeared later as a peer-reviewed publication in *Computers in Human Behavior*.

German University in Cairo

Sep 2021 – Jan 2023

Information Security Research Intern

- Applied security research with Prof. Wassim Alexan on cryptography and IoT/IoMT security. Published multiple papers on image encryption and secure medical data transmission.

TALKS & PRESENTATIONS

From Underground to Mainstream Marketplaces: Measuring AI-Enabled NSFW Deepfakes on

Feb 2026

Fiverr

USEC 2026, co-located with NDSS • San Diego, CA

Co-founder & Lead Maintainer, Egyptians in CS Research

2026

Community directory of 262 Egyptian computer science researchers across 16 research tracks worldwide. Expanded scope from AI to all of CS (security, systems, theory, HCI, and more) in collaboration with the original creator Badr AlKhamissi.

AAAS CASE Workshop Delegate

2025

One of four graduate students selected to represent Dartmouth College at the AAAS *Catalyzing Advocacy in Science and Engineering* Workshop; engaged with policymakers on science, technology, and public policy. Washington, D.C.

Conference Volunteer

2023

IEEE Secure Development Conference (IEEE SecDev 2023), Atlanta, GA. Supported session logistics and attendee coordination.

PhD Application Mentorship

Ongoing

Mentor a select cohort of prospective PhD applicants in Computer Science and related fields each application cycle, inspired by the guidance received during my own application journey. Provide personalized feedback on application materials (statements of purpose, research statements, CVs), help define research focus and target programs, and advise on overall application strategy and preparation.

Dartmouth Guarini School Student Spotlight

2025

Featured by Dartmouth's Guarini School of Graduate and Advanced Studies, discussing research in human-centered security and privacy, the journey from Zefta, Egypt to Dartmouth, and advocacy for international students in STEM policy.